10 (a) Write down the value of $100^{\frac{1}{2}}$

.....

(1)

(b) Find the value of $125^{\frac{2}{3}}$

.....

(2)

(Total for Question 10 is 3 marks)

11 3 teas and 2 coffees have a total cost of £7.80
5 teas and 4 coffees have a total cost of £14.20

Work out the cost of one tea and the cost of one coffee.

Let t and c be the prices of a tea and a coffee, in pounds.

(1) $3t + 2c = 7.80$

(2) $5t + 4c = 14.20$

1 mark
Form both equations

Double (1): $6t + 4c = 15.60$

Subtract (2): $t = 15.60 - 14.20 = 1.40$

1 mark
Eliminate c by subtracting equal coefficients

Substitute into (1): $3 \times 1.40 + 2c = 7.80$

1 mark
Substitute and solve for c

$$2c = 3.60$$
$$\div 2 \qquad \div 2$$
$$c = 1.80$$

tea £ 1.40

coffee £ 1.80

1 mark
Tea £1.40; coffee £1.80

(Total for Question 11 is 4 marks)

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